

Bandbreite der Programmiersprachen

Carsten Gips (HSBI)

Unless otherwise noted, this work is licensed under CC BY-SA 4.0.

99 Bottles of Beer

99 bottles of beer on the wall, 99 bottles of beer. Take one down and pass it around, 98 bottles of beer on the wall.

98 bottles of beer on the wall, 98 bottles of beer. Take one down and pass it around, 97 bottles of beer on the wall.

[...]

2 bottles of beer on the wall, 2 bottles of beer. Take one down and pass it around, 1 bottle of beer on the wall.

1 bottle of beer on the wall, 1 bottle of beer. Take one down and pass it around, no more bottles of beer on the wall.

No more bottles of beer on the wall, no more bottles of beer. Go to the store and buy some more, 99 bottles of beer on the wall.

Imperativ, Hardwarenah: C

```
#define MAXBEER (99)
void chug(int beers);
main() {
    register beers;
    for(beers = MAXBEER; beers; chug(beers--)) puts("");
    puts("\nTime to buy more beer!\n");
}
void chug(register beers) {
    char howmany[8], *s;
    s = beers != 1 ? "s" : "";
    printf("%d bottle%s of beer on the wall,\n", beers, s);
    printf("%d bottle%s of beeeer . . . ,\n", beers, s);
    printf("Take one down, pass it around,\n");
    if(--beers) sprintf(howmany, "%d", beers); else strcpy(howmany, "No more");
    s = beers != 1 ? "s" : "";
    printf("%s bottle%s of beer on the wall.\n", howmany, s);
}
```

Imperativ, Objektorientiert: Java

```
class bottles {  
    public static void main(String args[]) {  
        String s = "s";  
        for (int beers=99; beers>-1;) {  
            System.out.print(beers + " bottle" + s + " of beer on the wall, ");  
            System.out.println(beers + " bottle" + s + " of beer, ");  
            if (beers==0) {  
                System.out.print("Go to the store, buy some more, ");  
                System.out.println("99 bottles of beer on the wall.\n");  
                System.exit(0);  
            } else  
                System.out.print("Take one down, pass it around, ");  
            s = (--beers == 1)?"": "s";  
            System.out.println(beers + " bottle" + s + " of beer on the wall.\n");  
        }  
    }  
}
```

Logisch: Prolog

```
bottles :-  
    bottles(99).  
  
bottles(1) :-  
    write('1 bottle of beer on the wall, 1 bottle of beer, '), nl,  
    write('Take one down, and pass it around, '), nl,  
    write('Now they are all gone. '), nl, !.  
  
bottles(X) :-  
    write(X), write(' bottles of beer on the wall, '), nl,  
    write(X), write(' bottles of beer, '), nl,  
    write('Take one down and pass it around, '), nl,  
    NX is X - 1,  
    write(NX), write(' bottles of beer on the wall. '), nl, nl,  
    bottles(NX).
```

Funktional: Haskell

```
bottles 0 = "no more bottles"
bottles 1 = "1 bottle"
bottles n = show n ++ " bottles"

verse 0  = "No more bottles of beer on the wall, no more bottles of beer.\n"
        ++ "Go to the store and buy some more, 99 bottles of beer on the wall."

verse n  = bottles n ++ " of beer on the wall, " ++ bottles n ++ " of beer.\n"
        ++ "Take one down and pass it around, " ++ bottles (n-1)
                                                ++ " of beer on the wall.\n"

main     = mapM (putStrLn . verse) [99,98..0]
```

Quelle: "Language Haskell" by Iavor on 99-bottles-of-beer.net


```
fun fib(x) {  
  if (x == 0) {  
    return 0;  
  } else {  
    if (x == 1) {  
      return 1;  
    } else {  
      fib(x - 1) + fib(x - 2);  
    }  
  }  
}  
  
var wuppie = fib;  
wuppie(4);
```

- Compiler übersetzen formalen Text in ein anderes Format
- Berücksichtigung von unterschiedlichen
 - Sprachkonzepten (Programmierparadigmen)
 - Typ-Systemen
 - Speicherverwaltungsstrategien
 - Abarbeitungsstrategien



Unless otherwise noted, this work is licensed under CC BY-SA 4.0.

Exceptions:

- “Language Haskell” by Iavor on 99-bottles-of-beer.net
- “Language Java” by Sean Russell on 99-bottles-of-beer.net
- Abzählreim “99 Bottles of Beer” nach “Lyrics of the song 99 Bottles of Beer” on 99-bottles-of-beer.net
- “Language C” by Bill Wein on 99-bottles-of-beer.net
- “Language Prolog” by M@ on 99-bottles-of-beer.net
- Screenshot of “Language Brainfuck” by Michał Wojciech Tarnowski on 99-bottles-of-beer.net

